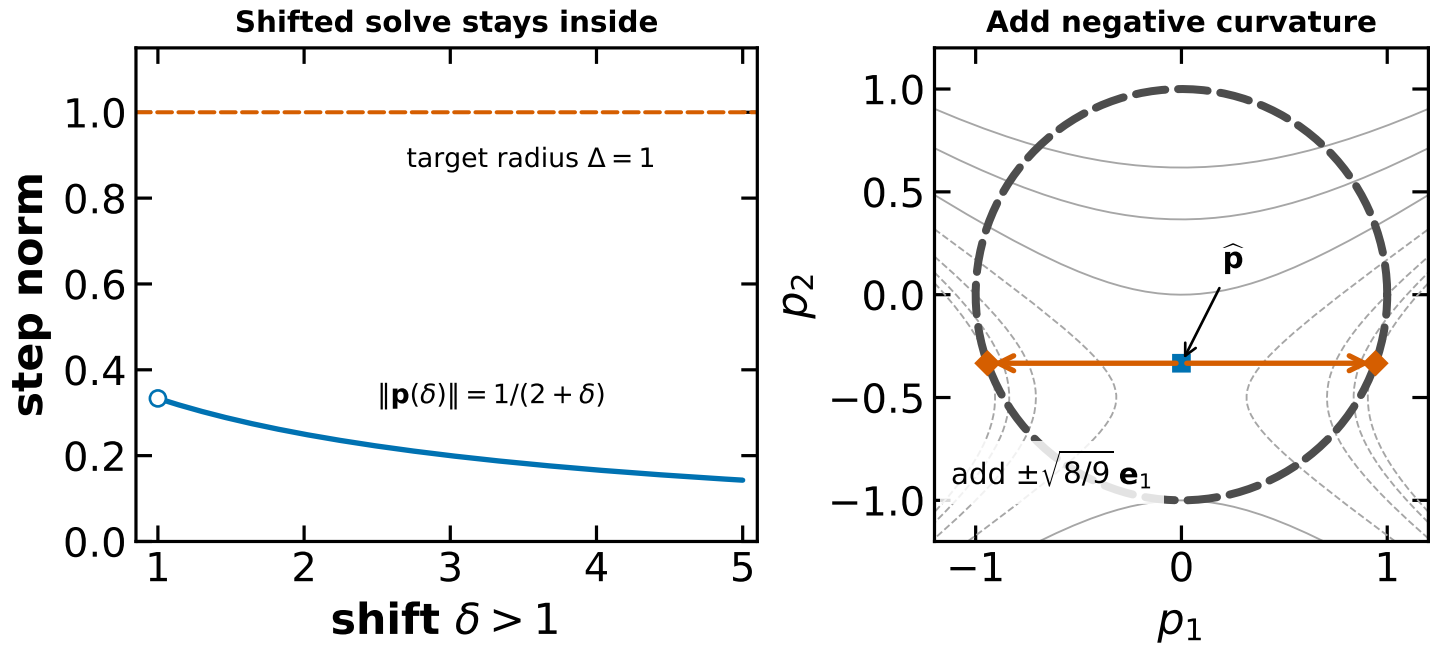
Shifted solve stays inside
step norm
1.0
0.8
0.6
0.4
0.2
0.0
target radius $\Delta = 1$
$\|\mathbf{p}(\delta)\| = 1/(2+\delta)$
1
2
3
4
5
shift $\delta > 1$
Add negative curvature
p_2
1.0
0.5
0.0
−0.5
−1.0
$\widehat{\mathbf{p}}$
add $\pm\sqrt{8/9}\,\mathbf{e}_1$
−1
0
1
p_1